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APPLIED RESEARCH

A Portable, Generalizable Machine Learning Framework for Long-Term Student Dropout Prediction

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ABSTRACT Student dropout undermines both student success and institutional performance. In recent years, various machine-learning models have been developed to predict dropout risk in higher education; however, existing studies remain constrained by four major gaps: (i) limited prediction horizons, (ii) first-year focus, (iii) inadequate handling of class imbalance, and (iv) lack of calibrated interpretability. To overcome these limitations, this study proposes a portable, generalizable machine-learning framework for long-term dropout prediction that systematically integrates modular components addressing each gap. Specifically, (i) rolling multi-semester windows generate horizon-specific targets several semesters in advance, (ii) an all-cohort student–semester dataset eliminates the first-year restriction by representing every enrolled cohort, (iii) a resampling–threshold optimization coupling mitigates class imbalance to improve sensitivity to rare events, and (iv) probability calibration with SHAP (SHapley Additive exPlanations)-based interpretability ensures reliable and transparent model outputs. Empirical evaluation on a university registry dataset demonstrates that the framework produces well-calibrated, temporally stable, and interpretable dropout-risk estimates. Because it relies solely on ubiquitous registry variables and off-the-shelf algorithms, the framework is designed to be portable and generalizable across higher-education institutions and suitable for integration into early-warning systems that support timely, data-driven student advising.

INDEX TERMS Class imbalance, early warning systems, explainability, higher education, machine learning, rolling time-window dataset, student dropout prediction, threshold optimization.

I. INTRODUCTION

Student dropout remains a persistent challenge for higher education institutions (HEIs) worldwide, and dropout rates are widely used as indicators of educational outcomes and institutional social responsibility [1], [2]. In OECD countries, on average, 12% of full-time undergraduates leave before the beginning of their second year; this figure rises to 21% by the end of the program's theoretical duration and to 23% three years later [3]. Dropout imposes substantial costs on students, HEIs, and society: individuals forfeit time, effort, and financial resources [4]; low retention can signal that an institution is not fulfilling its educational mission [5], damaging reputation and further depressing retention; and inadequate preparation for the workforce erodes human capital [6].

A large body of research investigates the drivers of continued enrollment [7], [8], yet converting such knowledge into evidence-based practices that measurably improve retention remains difficult [9]. A complementary, practice-oriented approach is to detect at-risk students early with data-driven prediction, thereby enabling targeted interventions and support programs [10], [11]. As machine learning gains traction in the social sciences, HEI decision-support applications have proliferated [12], [13], with student-dropout prediction emerging as a central focus [14], [15], [16], [17].

Differences in feature sets, sample sizes, cohort composition, and label definitions make existing dropout-prediction studies heterogeneous and shape both their objectives and practical reach. First, many studies focus exclusively on

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first-year students, since this group typically exhibits the highest dropout rates [7], [8], [18], [19], [20]. However, substantial attrition also occurs in later academic years [21], and restricting analysis to first-year students prevents assessment of temporal stability and limits model applicability to institution-wide retention strategies. Second, even when models include students across all academic years [22], [23], they often rely on features recorded close to the dropout event, limiting their usefulness for proactive, long-term intervention. In this study, we use the term ''long-term'' in a practical sense to refer to predicting dropout at least one semester in advance, rather than focusing solely on immediate withdrawal. The literature emphasizes that earlier identification enables greater prevention potential [10], leaving a need for frameworks that support early prediction several semesters in advance using universally available registry data.

Third, class imbalance remains a major challenge because dropout cases are far rarer than student retention. Models trained on imbalanced data tend to favor the majority class, and naive accuracy metrics can yield misleading conclusions [24], [25], [26]. Robust prediction therefore requires suitable evaluation metrics together with techniques such as data balancing and decision-threshold optimization, yet many studies under-address these issues. Finally, few works systematically consider probability calibration and explainability. Even highly accurate models may produce poorly calibrated probabilities or opaque decision logic, which undermines trust and adoption of predictive analytics in academic advising [27].

To address these gaps, we propose a comprehensive, generalized machine-learning framework for long-term dropout prediction. The goal is not to tailor a model to a single institution, but to provide a portable, end-to-end pipeline that any HEI can adopt with minimal customization. Here, ''generalized'' denotes design-level universality: each stage uses only ubiquitous registry variables (e.g., cumulative credits, recent GPA, enrollment status, years since admission) and off-the-shelf components (threshold tuning, calibration, SHAP). Multi-institution replication will further validate external performance; however, portability of the procedure does not depend on such replication. The framework thus serves as a practical guideline for building and adapting early-warning systems under local constraints.

Specifically, our framework (i) targets students across all academic years; (ii) uses only readily available academic data; (iii) predicts dropout risk several semesters in advance via rolling time-window datasets; and (iv) integrates class-imbalance handling, probability calibration, and SHAP-based explanations to produce reliable and interpretable alerts. It systematically covers dataset preparation, metric selection, label definition, model training, class-imbalance handling, evaluation, calibration, and explainability. We compare multiple classifiers, optimize the decision threshold, and apply light data-balancing methods to obtain robust, practitioner-ready results. By combining probability calibration with SHAP-based explanations, the framework improves both reliability and interpretability, enabling actionable early interventions across diverse higher-education contexts. Built entirely on ubiquitous registry data and standard algorithms, it is inherently portable across institutions and can be adopted with minimal local adaptation.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed framework and its stage-wise empirical results. Section IV discusses the findings, including comparisons with recent full-pipeline studies. Section V concludes and outlines directions for future research.

II. RELATED WORK

To address the four research gaps outlined in Section I, we review prior work along three axes: (i) student cohort and prediction horizon; (ii) strategies for imbalanced data; and (iii) model calibration and explainability.

A substantial portion of machine-learning studies on dropout prediction concentrates on first-year students, reflecting the high attrition observed early in the program. These studies also differ in the temporal gap between data collection and the eventual dropout event. Early-dropout prediction from freshman-year data includes work that uses first-year course grades to predict first-year dropout [28]; a model that relies solely on information available at admission to predict first-year outcomes [8]; and a two-semester-ahead model trained on first-semester data, including unstructured text [20]. First-year data have also been used to predict outcomes across the entire college career: one line of work built models for both first-year dropout and graduation/dropout within six years using data from 66,060 first-year students [29], and other studies developed long-term models trained on pre-study, decision, and early-study phases [30] or proposed random-forest–based methods trained on first-semester data from 41,098 students [19]; a multi-university study likewise used first-semester data to forecast pre-graduation dropout [7]. More recently, a three-class dropout-prediction model (Graduate, Dropout, Enrolled) was developed using data from a Portuguese university, but because the prediction horizon was not defined, its real-world applicability for early-warning purposes remains limited [31].

Table 1 summarizes first-year–focused studies. Despite variation in metrics and features, a broad pattern emerges: performance tends to decline as the prediction horizon lengthens. Many models also impose no minimum horizon—that is, they classify dropouts occurring immediately after data collection—thereby falling short of true long-term prediction. For actionable early-warning systems, institutions typically require at least one full semester of lead time; excluding near-instant dropouts would almost certainly reduce reported performance further, underscoring the need for models that maintain reliability several semesters in advance.

TABLE 1. Prediction horizons and results of models trained on first-year cohorts.

Ref.	Dataset size	Prediction horizon (data → dropout)	AUC / F1[a]
[28]	4,434	of first year → in first year	F1 0.73[b]
[29][c]	66,060	of first year → in first year	AUC 0.89
		of first year → in first six years	AUC 0.81
[8]	5,951	at admission → in first year	F1 0.37
[20]	14,391	of first semester → in next two semesters	AUC 0.80
[30]	17,910	of first three semesters → to graduation	AUC 0.86
[7]	50,095	of first semester → to graduation	AUC 0.78[b]
[19]	41,098	of first semester → to graduation	F1 0.83
[30]	17,910	at admission → to graduation	AUC 0.77

[a] AUC reported where available; otherwise, F1 is shown.
[b] Average across multiple models.
[c] Short- and long-term results reported separately.

Although first-year attrition is prominent, evidence indicates that a sizable share of dropouts occurs in subsequent years, yielding a relatively even distribution across academic years [23], [32], [33]. This motivates frameworks that cover all-year students [21]. Survival-analysis–based prediction has been explored with semester-wise characteristics up to six semesters, training separate models for each completed-semester count; however, splitting by semester reduces sample sizes per model and risks underfitting [22]. Another line aggregates semester-wise data to single values (means or most recent) to standardize input length [23]; yet using data immediately prior to dropout shortens the horizon and limits proactive intervention. In contrast, our rolling-window framework preserves temporal trajectories and provides fixed-horizon early-warning predictions, which are not supported by either survival-analysis models or aggregation-based approaches. More recent studies have attempted all-year, long-term prediction [34], [35], but the inclusion of near-instant dropouts prevents these models from being truly long-term in the operational sense.

Class imbalance represents a second, recurring limitation. When dropout cases are far rarer than retention, classifiers trained on imbalanced data tend to favor the majority class, and metric choice can mask this bias [24], [25], [26]. Robust evaluation thus calls for appropriate metrics and techniques such as light resampling and decision-threshold optimization. A systematic review reports that only 21 of 100 surveyed studies explicitly accounted for imbalance [16]. Data-level balancing (oversampling/undersampling) appears frequently [7], [23], [28], [35], [36], and balanced-sampling schemes tailored to dropout have been compared against numerous baselines with gradient-boosting models [32]. At the same time, evidence suggests that strong learners (e.g., XGBoost, CatBoost) may rely effectively on threshold optimization without SMOTE-like balancing [37]. Threshold selection is crucial for navigating the recall–precision trade-off, yet relatively few studies treat it as a first-class design choice [19], [34]. For imbalanced datasets, the precision–recall (PR) curve is generally more informative than the receiver operating characteristic (ROC) curve [24], although many works still report AUC-ROC rather than average precision (AP/AUC-PR).

Calibration and explainability form the third axis. Calibration aligns predicted probabilities with empirical frequencies, while explainability clarifies how and why a prediction is produced. Both properties are central to practitioner trust, yet remain underrepresented in education-domain studies [27]. Explicit applications of probability calibration in dropout prediction are rare; one example demonstrates calibration alongside predictive modeling [38]. Further, researchers have highlighted the scarcity of calibration and explainability and illustrated the use of post-hoc explanations across K–12 and higher-education datasets [10], [39]. Together, the literature confirms the four gaps identified in Section I—limited prediction horizon, cohort restriction, inadequate handling of class imbalance, and scarce attention to calibration and explainability. To clarify how our study departs from these existing works, Table 2 summarizes the key differences in research objectives and methodological focus between traditional institution-specific approaches and our proposed generalizable framework. Guided by this design philosophy, Section III details the proposed comprehensive, end-to-end pipeline.

III. PROPOSED FRAMEWORK

The proposed pipeline addresses, step by step, the four research gaps identified earlier. Before presenting technical details, this section (i) states the guiding design principles, (ii) presents Table 3 as a five-stage roadmap, and (iii) provides a high-level tour of each stage. To ensure transparency and modularity, the framework adopts a multi-stage structure in which each step performs a conceptually distinct operation that institutions can reuse or replace independently. In addition, separating Stages 3–5 allows candidate models to be progressively narrowed before applying cost-intensive procedures. Implementation details—algorithms, hyperparameters, and case-study figures—appear in Sections III-A–III-E.

Design principles:

1) **Cohort-wide coverage** — rely exclusively on registry data and design the pipeline to cover *all academic years*, eliminating the common first-year-only restriction.
2) **Long-term early warning** — guarantee a non-zero minimum prediction horizon via rolling t-semester windows, directly addressing the limited-horizon gap.
3) **Imbalance awareness** — combine algorithm-level techniques such as decision-threshold optimization with data-level techniques such as resampling, so skewed class distributions are handled explicitly rather than masked by overall accuracy.
4) **Actionability** — provide calibrated probabilities and SHAP-based explanations, enabling advisors to both *trust* (calibration) and *understand* (explainability) each

TABLE 2. Comparison of research objectives and methodological focus: Institution-specific studies vs. the proposed generalized framework.

Comparison dimension	Institution-specific approaches (Existing studies)	Proposed framework (This study)
Primary objective	Site-specific model development (Build a high-performing model for a specific dataset)	Generalizable methodology proposal (Establish a standard pipeline for *any* institution)
Methodological focus	Result-Centric (Focus on maximizing accuracy metrics)	Process-Centric (Focus on optimizing the entire pipeline)
Completeness of pipeline	Implicit / Partial (Key implementation details often omitted)	Explicit / End-to-end (Fully specified for reproducibility and adaptation)
Portability	Low (Context-dependent) (Difficult to replicate without identical data schema)	High (Design-level universality) (Adaptable guidelines using ubiquitous registry data)

TABLE 3. Overview of the proposed framework.

Stage	Main objective(s)	Key methods or tasks	Gap(s) addressed
1. Data preparation	Integrate multi-semester registry; ensure data quality	Data ingestion; cleaning; basic transformation	– (foundation)
2. Sample construction	Generate time-window samples and dropout labels	Rolling-window labeling; t-semester horizon; remove ambiguous samples	limited horizon; cohort restriction
3. Base modeling	Train and tune candidate models on original data; optimize threshold	Encoding & scaling; CatBoost, XGBoost; cross-validation; F1-based threshold search	limited horizon; class imbalance
4. Data balancing	Mitigate class imbalance via resampling and retain the best model	Resample → retrain + re-optimize → compare with Stage 3	class imbalance
5. Post-hoc processing	Improve probability reliability and interpretability	Platt/isotonic scaling; SHAP explanations	calibration; explainability

alert, thereby closing the final gap—scarce attention to these issues.

Stage 1 ingests consecutive semester snapshots from the student information system and, after cleaning, feature selection, and basic transformation, merges them into a quality-controlled longitudinal registry. Stage 2 converts the registry into training *samples* by sliding a window across each student timeline, thus covering all academic years and ensuring sufficient lead time. Stage 3 evaluates a pool of classifiers—including state-of-the-art algorithms such as CatBoost and XGBoost—under the original class distribution, tunes hyperparameters by cross-validation, and selects an F1-maximizing decision threshold as an algorithm-level remedy for class imbalance. Stage 4 retrains the top-performing algorithms on multiple resampled variants of the training set (SMOTE, ADASYN, ENN, Tomek-links, etc.), re-optimizes the threshold, and retains whichever model—the original or a resampled one—yields the higher validation F1 or MCC. Finally, Stage 5 calibrates the chosen model—via Platt scaling or isotonic scaling on a held-out split—and computes SHAP values, so advisors can understand why each student is flagged. The next subsections apply the framework to the university dataset used in this study, thereby illustrating each stage in concrete detail.

A. DATA PREPARATION

This empirical study uses only those institutional records that every university already keeps—demographics, enrollment status, and semester grades—making the example implementation readily reproducible across most HEI contexts. The framework itself is, however, *feature-agnostic*: institutions may enrich the core registry with survey responses, LMS (learning management system) activity logs, or other local sources without modifying downstream stages. Source data are released once per semester on predetermined snapshot dates, and each row in the resulting table represents one student–semester pair.

The study analyzes 81,102 anonymized records from a South Korean university that operates a two-semester system (Spring: March–June; Fall: September–December) covering 2021–2023. All data were fully de-identified before analysis. Snapshots for the Spring semester are released on 1 October and those for the Fall semester on 1 April of the following year. To ensure data quality and maximize portability, we retained only features with no missing values; first-semester entrants and transfer students were excluded because their academic-performance fields are necessarily missing. This exclusion defines the models valid prediction scope rather than a selection bias, as the framework targets students with at least one semester of academic history from which academic-performance indicators are derived. After cleaning, the dataset contains 74,073 rows. Additional label-specific filtering is described in Section III-B.

Table 4 lists the retained features. Because the registry is rebuilt every semester, a student can appear multiple times with updated dynamic attributes such as gpa_last

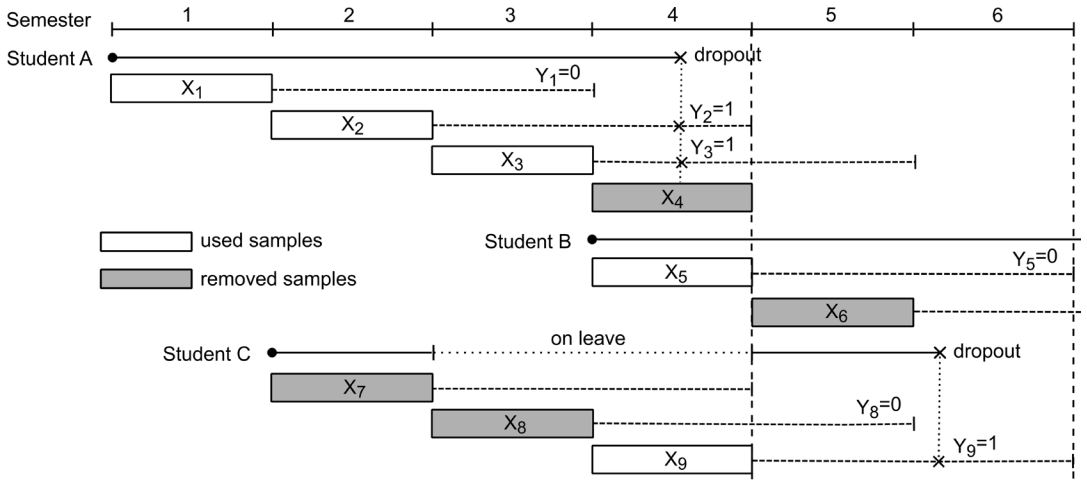


FIGURE 1. Rolling two-semester windows ($t = 2$) with used/removed samples and labels for three students.

TABLE 4. Dataset features.

Feature	Description	Type[a]
Static:		
release	Snapshot: spring or fall	Cat.
sex	Sex: male or female	Cat.
nation	Nationality (6 groups: Korea, Japan, Vietnam, …)	Cat.
adm_type	Type of admission (5 types)	Cat.
in_quota	In-quota or out-of-quota admission	Cat.
Dynamic:		
college	Affiliated college (6 colleges)	Cat.
year	Academic year (1–5)	Num.
years_since	Calendar years since initial admission (including leave periods)	Num.
gpa_last	GPA in the last semester (0–4.5)	Num.
credits_last	Credits earned in the last semester	Num.
credits_tot	Cumulative credits earned	Num.
n_semesters	Number of semesters enrolled	Num.
status	Academic status in the snapshot (enrolled / graduated / dropped out / on leave)	Cat.

[a]Cat. = Categorical, Num. = Numerical.

or credits_tot. We treat each semester-specific record as an independent training sample; this choice departs from first-year-only datasets in earlier work and yields far more observations, which in turn improve predictive performance. Dropout status includes both formal withdrawals and implicit non-registrations. Students who return within the approved leave duration are classified as enrolled, whereas those remaining unregistered after the authorized leave period are classified as dropout. All status changes are reflected in the subsequent semester snapshot.

Although the core dataset relies solely on universally available registry variables to ensure portability, the framework is extensible so that each institution can augment these with locally available variables to develop customized models using the same end-to-end pipeline.

B. SAMPLE CONSTRUCTION

For each record X, let y_i denote whether the student drops out in the i-th subsequent semester (1 = dropout, 0 = retention). A sample observed in Spring 2021, for instance, obtains y_1 by looking up the same student's status in the Fall 2021 snapshot. The task label is then defined as

$$Y(t) = \begin{cases} 0, & \text{if } \sum_{i=1}^{t} y_i = 0, \\ 1, & \text{otherwise,} \end{cases}$$

where t is the prediction horizon. Section III-C first compares models for $t \in \{1, 2, 3\}$ and then focuses subsequent analyses on $t = 2$.

Figure 1 illustrates rolling two-semester windows ($t = 2$) for three example students. Each horizontal bar is the window's feature vector X. A dotted line extending two semesters to the right visualizes the prediction horizon, and the corresponding label Y is printed just above that line. Windows drawn in white are kept in the dataset, whereas shaded ones are removed.

For illustration, consider Student A, who matriculates in Semester 1 and drops out in Semester 4. The rolling procedure therefore generates four candidate windows X_1–X_4. Because no dropout occurs within the two-semester horizon of X_1, its label $Y_1 = 0$; by contrast, Y_2 and Y_3 are set to 1 because the dropout falls inside their horizons. Any window whose status in that snapshot is already *dropped out* or *graduated* (e.g., X_4) is excluded, since future prediction would be meaningless. Student B illustrates a different filter: the horizon of X_6 extends beyond the last available snapshot, so the label cannot be determined, and the window is removed. More generally, the final t semesters of every student's timeline are removed whenever the full t-semester horizon cannot be observed.

In this study, a student is labeled *retention* when the current status is either *enrolled* or *graduated*. Conversely, the state *on leave* is deemed ambiguous and is excluded from model training. Window X_7 of Student C (Figure 1) exemplifies this case; any window that remains on leave throughout the entire t-semester horizon is discarded. By contrast, windows such as X_8 and X_9 are *initially* retained because, although the student

is currently on leave, the dropout label can still be resolved within the horizon.

Leave periods require special handling—especially in South Korea, where male students often take a two-year military leave. Successive windows from the same leave spell (e.g., X_8 and X_9) are identical except for years_since. Such near-duplicates can be problematic: if they carry different labels, the model receives conflicting signals; if they carry the same label and end up split between training and test sets, performance is artificially inflated. To avoid both issues, we retain only the *last* window of each leave segment (e.g., X_9).

After all filtering, the working datasets contain 38,814, 31,769, and 23,021 windows for $t = 1, 2, 3$, with positive (dropout) rates of 0.034, 0.051, and 0.053, respectively. The final dataset was partitioned into a stratified 80% training split and 20% test split, and all model fitting, tuning, and threshold optimization were performed exclusively on the training split; the test split was held out for final evaluation only.

C. BASE MODELING

Given that our dataset consists of structured, semester-level registry records rather than high-frequency behavioral logs, we prioritized representative tabular classifiers that are computationally efficient and robust to heterogeneous administrative features. This selection aligns with our goal of ensuring operational portability across standard institutional environments. Specifically, we evaluated six models: support vector machine (SVM), multilayer perceptron (MLP), LightGBM, XGBoost, CatBoost, and a stacking ensemble that uses those five base learners and logistic regression as the meta-learner. SVM is a margin-based classifier with an RBF kernel, and MLP is a fully connected feed-forward neural network. LightGBM, XGBoost, and CatBoost are gradient-boosted tree methods that remain the most widely adopted and empirically validated algorithms for structured tabular data, consistently outperforming or matching deep-learning architectures while offering higher stability and lower risk of overfitting with moderate-sized datasets. The proposed framework is model-agnostic and can readily incorporate newer classifiers if desired. All experiments were conducted in Python 3.12 with scikit-learn 1.4, lightgbm 4.5, xgboost 2.0, and catboost 1.2.

Table 5 summarizes the confusion-matrix layout and the metrics derived from it. Rows are true classes, columns predicted classes, and each cell gives the sample count (e.g., *TP* is the number of correctly predicted dropouts). Accuracy and specificity are uninformative for our highly imbalanced data (positive rate < 0.06); therefore we focus on recall, precision, and their harmonic mean F1. We also report the area under the precision–recall curve (average precision, AP—more diagnostic than AUC-ROC on skewed data [40], [41]), the area under the ROC curve (AUC-ROC), and the Matthews correlation coefficient (MCC—a balanced measure even under severe class imbalance [42]):

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}.$$

TABLE 5. Confusion matrix and performance metrics.

		Predicted class		Metrics
		$\hat{Y}=0$	$\hat{Y}=1$	
True class	$Y=0$	TN (True Negative)	FP (False Positive)	Specificity = TN / (TN+FP)
	$Y=1$	FN (False Negative)	TP (True Positive)	Recall = TP / (TP+FN)
Metrics			Precision = TP / (TP+FP)	Accuracy = (TN+TP) / N [a]

[a] Here N = TN + TP + FN + FP.

TABLE 6. Hyperparameter grid and categorical-encoding scheme.

Classifier	Hyperparameter and search space	Encod.
SVM	kernel: {'rbf'}, C: {0.1,1,10}, gamma: {0.01,0.1,1}	One-hot
MLP	activation: {'relu'}, hidden_layer_sizes: {(10,10,10), (12,12,12)}	One-hot
XGBoost	n_estimators: {40, ···, 110}, max_depth: {4, ···, 10}	Label
LightGBM	n_estimators: {40, ···, 110}, max_depth: {4, ···, 10}	Label
CatBoost	iterations: {800, 1000}, depth: {6, 7, . . ., 10}	Built-in
Stacking	final_estimator: {LogisticRegression}, C: {1, 10}	One-hot

Regarding feature preprocessing, numerical features were standardized to zero mean and unit variance. CatBoost employed its native ordered target encoding, whereas the other models compared one-hot and label encodings, adopting the better-performing scheme based on cross-validation results.

To ensure methodological reproducibility, we employed a systematic grid search with 5-fold cross-validation, as summarized in Table 6. For the primary gradient-boosting models (XGBoost, LightGBM, CatBoost), we tuned key structural hyperparameters—specifically searching the number of estimators and maximum depth within the ranges specified in Table 6—as these ranges were identified as effective regions during preliminary trials on our registry data. Similarly, for non-tree baselines, we optimized parameters such as C and γ for SVM, and the hidden-layer architecture for MLP. Any hyperparameters not explicitly listed in Table 6 were fixed to their library defaults. While optimal configurations may vary across institutional datasets, this standardized protocol ensures that the model selection process remains consistent and transparent. Logistic regression was used as the stacking meta-learner to maintain simplicity and prevent overfitting. Because the study aims to validate the framework rather than to rank classifiers statistically, we report descriptive metrics without conducting significance tests between models.

Most classifiers return either probabilities (threshold 0.5 by default) or decision scores (threshold 0 by default);

TABLE 7. Evaluation on test set.

t	Classifier	AP	AUC	Threshold = 0.5				Threshold optimization			
				Recall	Precision	F1	MCC [a]	Recall	Precision	F1	MCC
1	SVM	0.414	0.853	0.229	0.720	0.348	0.395	0.476	0.454	0.465	0.444
	MLP	0.455	0.929	0.301	0.642	0.410	0.426	0.551	0.448	0.495*	0.476*
	LightGBM	0.492	0.931	0.318	0.650	0.428	0.441	0.579	0.413	0.482	0.466
	XGBoost	0.471	0.927	0.291	0.639	0.400	0.417	0.575	0.434	0.495*	0.477*
	CatBoost	0.504*	0.936	0.298	0.696	0.417	0.443	0.568	0.438	0.495*	0.477*
	Stacking	0.510*	0.924	0.312	0.700	0.431	0.454	0.575	0.444	0.501*	0.484*
2	SVM	0.413	0.812	0.217	0.726	0.334	0.382	0.443	0.453	0.448	0.420
	MLP	0.431	0.896	0.267	0.625	0.374	0.390	0.494	0.447	0.469	0.441
	LightGBM	0.465	0.904	0.289	0.643	0.399	0.413	0.519	0.426	0.468	0.440
	XGBoost	0.466	0.904	0.289	0.681	0.406	0.427	0.509	0.464	0.486*	0.458*
	CatBoost	0.484*	0.911	0.277	0.715	0.399	0.429	0.494	0.480	0.487*	0.459*
	Stacking	0.483*	0.900	0.296	0.667	0.410	0.426	0.509	0.485	0.497*	0.470*
3	SVM	0.381	0.798	0.211	0.684	0.322	0.363	0.368	0.500	0.424	0.402
	MLP	0.375	0.870	0.259	0.587	0.360	0.369	0.409	0.463	0.434	0.406
	LightGBM	0.450	0.897	0.243	0.698	0.360	0.394	0.449	0.455	0.452	0.422
	XGBoost	0.437	0.898	0.215	0.646	0.322	0.354	0.449	0.481	0.464*	0.436*
	CatBoost	0.464*	0.900	0.223	0.679	0.335	0.372	0.433	0.482	0.456*	0.428*
	Stacking	0.466*	0.887	0.255	0.677	0.371	0.398	0.474	0.440	0.456*	0.425*

[a]MCC ranges from –1 (perfectly wrong) through 0 (random) to 1 (perfect).
An asterisk (*) denotes the best values for each horizon.

TABLE 8. Robustness test under temporal splits ($t = 2$).

Classifier	AP	AUC	F1(opt.)	MCC(opt.)
LightGBM	0.472 (0.465)	0.894 (0.904)	0.483 (0.468)	0.452 (0.440)
XGBoost	0.466 (0.466)	0.894 (0.904)	0.472 (0.486)	0.447 (0.458)
CatBoost	0.481 (0.484)	0.897 (0.911)	0.493 (0.487)	0.468 (0.459)
Stacking	0.495 (0.483)	0.885 (0.900)	0.497 (0.497)	0.464 (0.470)

Values in parentheses are the results from the 80/20 random split (Table 7).

TABLE 9. Resampling techniques and the resulting positive rate for $t = 2$.

Type	Technique	Brief strategy description	Pos. rate
Over-sampling	SMOTE	add synthetic minority samples	0.500
	ADASYN	focus on hard minority samples	0.501
Under-sampling	CNN	remove redundant majority samples	0.322
	ENN	remove inconsistent majority samples	0.056
	Tomek-links	remove ambiguous majority samples	0.053
Hybrid	SMOTE–ENN	SMOTE followed by ENN	0.521

we classify a sample as positive when it exceeds the corresponding threshold. Because lowering the threshold raises recall and lowers precision, we sweep it from 0 to 1 on five-fold cross-validated predictions and pick the value that maximizes F1. Threshold tuning for resampled models is revisited in Section III-D.

Table 7 reports test-set scores for the three prediction horizons ($t = 1, 2, 3$). SVM performs worst overall. MLP is competitive at $t = 1$ but degrades as the horizon lengthens. CatBoost and Stacking yield the highest AP, while XGBoost attains the best F1 at $t = 3$ after threshold tuning. CatBoost's AUC ranges from 0.900 to 0.936, surpassing the best results reported in previous dropout-prediction studies (Table 1). Stacking adds considerable training cost for marginal gains, so CatBoost offers the best cost–performance trade-off. The marginal improvement from stacking likely arises from the strong, correlated performance of base learners such as CatBoost and XGBoost, leaving limited residual variance for the meta-learner to capture.

Optimizing the threshold boosts F1 substantially; for $t = 2$, CatBoost's F1 rises from 0.399 to 0.487 (+22%). As expected, the optimized F1 declines as the horizon grows, but the drop from $t = 1$ to 2 is smaller than the drop from $t = 2$ to 3. Moreover, a two-semester horizon ($t = 2$) still gives advisors ample time to intervene and prevent an at-risk student from dropping out. Therefore, we conduct all subsequent experiments with $t = 2$, which offers the best balance between lead time and accuracy. Operationally, this F1 improvement indicates that more at-risk students can be identified early without a disproportionate increase in false-positive alerts, making the resulting early-warning signals more actionable for advisors.

In practical deployment settings, dropout prediction models are trained on all historically available student records and used to forecast outcomes in future semesters. Under this scenario, temporal leakage—where information from later semesters unintentionally influences the training stage—is a legitimate concern. To examine whether such leakage affects our results, we performed an additional temporal-split experiment in which models are trained only on earlier semesters and evaluated on strictly later semesters.

A temporally valid split can be constructed for the $t = 2$ horizon. In our evaluation, we use windows from 2021–1 and 2021–2 as the training set, whose two-semester-ahead horizons lie in 2022–1 and 2022–2, and we use windows from 2022–2 as the test set, whose corresponding horizon lies in 2023–1 and 2023–2. These horizons do not overlap,

TABLE 10. Validation-test scores for $t = 2$ after each resampling method.

Classifier	Metric	Baseline	SMOTE	ADASYN	CNN	ENN	Tomek-links	SMOTE–ENN
SVM	AP	0.413	0.359	0.341	0.425*	0.403	0.421*	0.387
	F1(default)	0.334	0.378	0.332	0.444	0.458	0.393	0.355
	F1(opt.)	0.448	0.351	0.334	0.464*	0.444	0.474*	0.392
	MCC(default)	0.382	0.368	0.325	0.433	0.432	0.408	0.358
	MCC(opt.)	0.420	0.318	0.300	0.435*	0.415	0.448*	0.359
MLP	AP	0.431*	0.398	0.396	0.396	0.417	0.425*	0.362
	F1(default)	0.374	0.351	0.339	0.434	0.454	0.428	0.334
	F1(opt.)	0.469*	0.381	0.399	0.426	0.429	0.470*	0.387
	MCC(default)	0.390	0.352	0.339	0.405	0.425	0.424	0.339
	MCC(opt.)	0.441*	0.353	0.367	0.397	0.399	0.442*	0.355
LightGBM	AP	0.465*	0.443	0.420	0.434	0.467*	0.451	0.431
	F1(default)	0.399	0.458	0.433	0.452	0.449	0.425	0.458
	F1(opt.)	0.468	0.436	0.432	0.459	0.481*	0.475*	0.454
	MCC(default)	0.413	0.429	0.402	0.429	0.422	0.426	0.442
	MCC(opt.)	0.440	0.412	0.405	0.430	0.458*	0.450*	0.433
XGBoost	AP	0.466*	0.431	0.430	0.428	0.441	0.452*	0.419
	F1(default)	0.406	0.446	0.428	0.449	0.447	0.392	0.402
	F1(opt.)	0.486*	0.446	0.443	0.442	0.469	0.487*	0.442
	MCC(default)	0.427	0.421	0.404	0.426	0.420	0.397	0.391
	MCC(opt.)	0.458*	0.416	0.413	0.412	0.443	0.462*	0.417
CatBoost	AP	0.484*	0.449	0.443	0.443	0.467	0.490*	0.437
	F1(default)	0.399	0.454	0.449	0.442	0.489*	0.428	0.454
	F1(opt.)	0.487*	0.402	0.412	0.472	0.489*	0.477	0.420
	MCC(default)	0.429	0.433	0.429	0.425	0.465	0.438	0.430
	MCC(opt.)	0.459*	0.404	0.414	0.443	0.466*	0.452	0.407
Stacking	AP	0.483*	0.466	0.455	0.454	0.457	0.483*	0.448
	F1(default)	0.410	0.481	0.473	0.453	0.484	0.448	0.453
	F1(opt.)	0.497*	0.442	0.449	0.481	0.490	0.500*	0.441
	MCC(default)	0.426	0.454	0.444	0.442	0.458	0.451	0.435
	MCC(opt.)	0.470*	0.411	0.419	0.453	0.465	0.475*	0.411

An asterisk (*) denotes the best values for each model.

ensuring that the temporal order is preserved and that only past information is used during training.

The results in Table 8 show that classifier performance under this temporal split closely matches the scores obtained under the 80/20 random split, demonstrating that the original results are robust and not inflated by temporal leakage.

D. DATA BALANCING

Building on Section III-C, which showed that threshold optimization improves the original imbalanced models, we assess whether data-level remedies—oversampling and undersampling—provide additional gains. Six benchmark techniques (Table 9) were applied to the training split only and evaluated with imbalanced-learn 0.12.

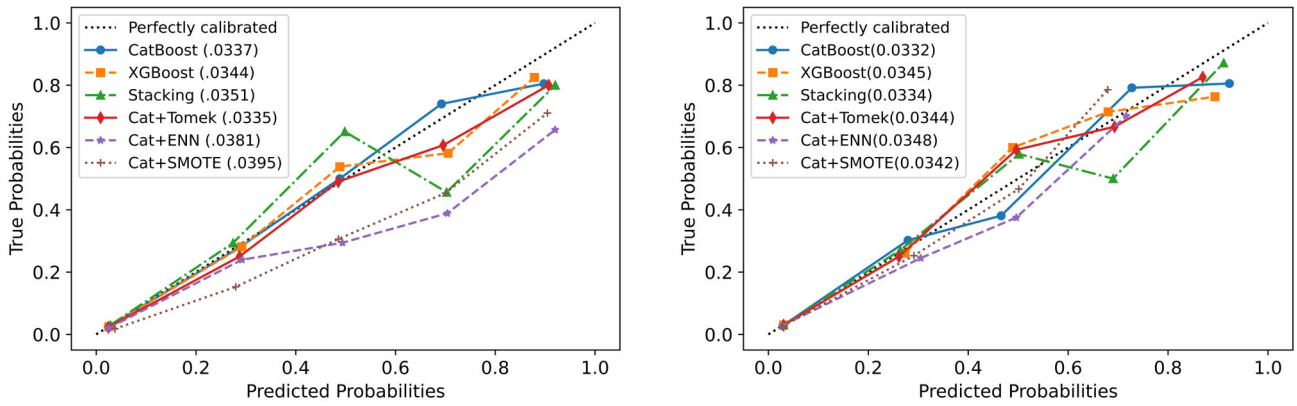
SMOTE (synthetic minority oversampling technique) and its variant ADASYN [43], [44] create synthetic minority rows, whereas CNN (condensed nearest neighbors), ENN (edited nearest neighbors), and Tomek-links [45], [46], [47] delete majority rows. SMOTE–ENN combines both steps [48]. Note that ENN and Tomek-links hardly change the original positive rate (0.051)—a point revisited below.

Because resampling alters the class distribution, performing cross-validation and threshold optimization on the resampled dataset itself can bias the tuned threshold and yield unreliable estimates. We therefore adopt a two-scenario protocol:

1) **No threshold optimization** — keep the original 80/20 train–test split, apply the chosen resampling method to the 80% training portion, fit the model once, and measure AP, F1(default), and MCC(default) on the untouched 20% test set;
2) **With threshold optimization** — carve out one quarter of the 80% training portion, yielding a 60/20/20 split (train/validation/test). After resampling the 60% train slice, fit the model, optimize the threshold on the 20% validation slice, and report F1(opt.) and MCC(opt.) on the held-out 20% test set.

Table 10 summarizes AP, F1, and MCC under each resampling method. Baseline rows are copied verbatim from Table 7 for side-by-side comparison with the non-resampled models. Undersampling (ENN, Tomek-links) outperformed oversampling across all classifiers. SMOTE–ENN narrowed the gap in F1 and MCC but still lagged in AP. ENN and Tomek-links require minimal data changes yet consistently deliver the best scores, indicating they are both efficient and effective.

Threshold re-optimization helped primarily when the resampling preserved a class ratio close to the original—most clearly with Tomek-links and, to a lesser extent,



(a) Before calibration (b) After Platt scaling

FIGURE 2. Calibration curves and Brier scores for selected models ($t = 2$). Curves closer to the diagonal indicate better calibration, and lower Brier scores are better (Cat = CatBoost; Cat + Tomek/ENN/SMOTE denote CatBoost models trained with Tomek-links, ENN, and SMOTE resampling, respectively).

with ENN. Because the validation set must retain the natural 0.05 positive rate, whereas oversamplers such as SMOTE, ADASYN, or SMOTE–ENN drive the training set toward 0.5, the resulting mismatch skews the tuned threshold and often *hurts* F1 and MCC.

Overall, Tomek-links or ENN slightly—but reliably—outperform the baseline across all classifiers, whereas SMOTE-type oversampling produced mixed results. For strong learners such as CatBoost or XGBoost, the marginal benefit of undersampling is small, but the computational cost is also low, making Tomek-links or ENN a low-cost, robust preprocessing choice.

E. POST-HOC PROCESSING

In many applications, the predicted probability itself—rather than the binary label—conveys a model's confidence. Saying that a student has an 80% chance of dropping out is more informative than simply stating "will drop out." A classifier is *calibrated* when the observed frequency of positives matches the predicted probability [24]. For example, if the model assigns a score of 0.8 to ten students, roughly eight would be expected to withdraw. If fewer than eight occur, the model is *overconfident*, whereas more implies *underconfident*. Calibration must therefore be verified—and, if necessary, adjusted—before probabilities can be interpreted as risk scores.

Figure 2 compares the calibration of six representative models: panel (a) shows raw outputs, whereas panel (b) shows the same models after Platt scaling [49]. Each curve plots the observed positive rate ("True Probabilities") against the predicted probability; the dotted diagonal marks perfect calibration. The number in each legend entry is the Brier score—the mean squared error of the predicted probabilities. The score ranges from 0 (perfect calibration) to 1 (worst), so smaller values indicate better calibration. While the Brier score provides a single summary value, the *shape* of the curve reveals whether errors are systematic (overconfidence or underconfidence) or merely noisy, so visual inspection complements the numeric score.

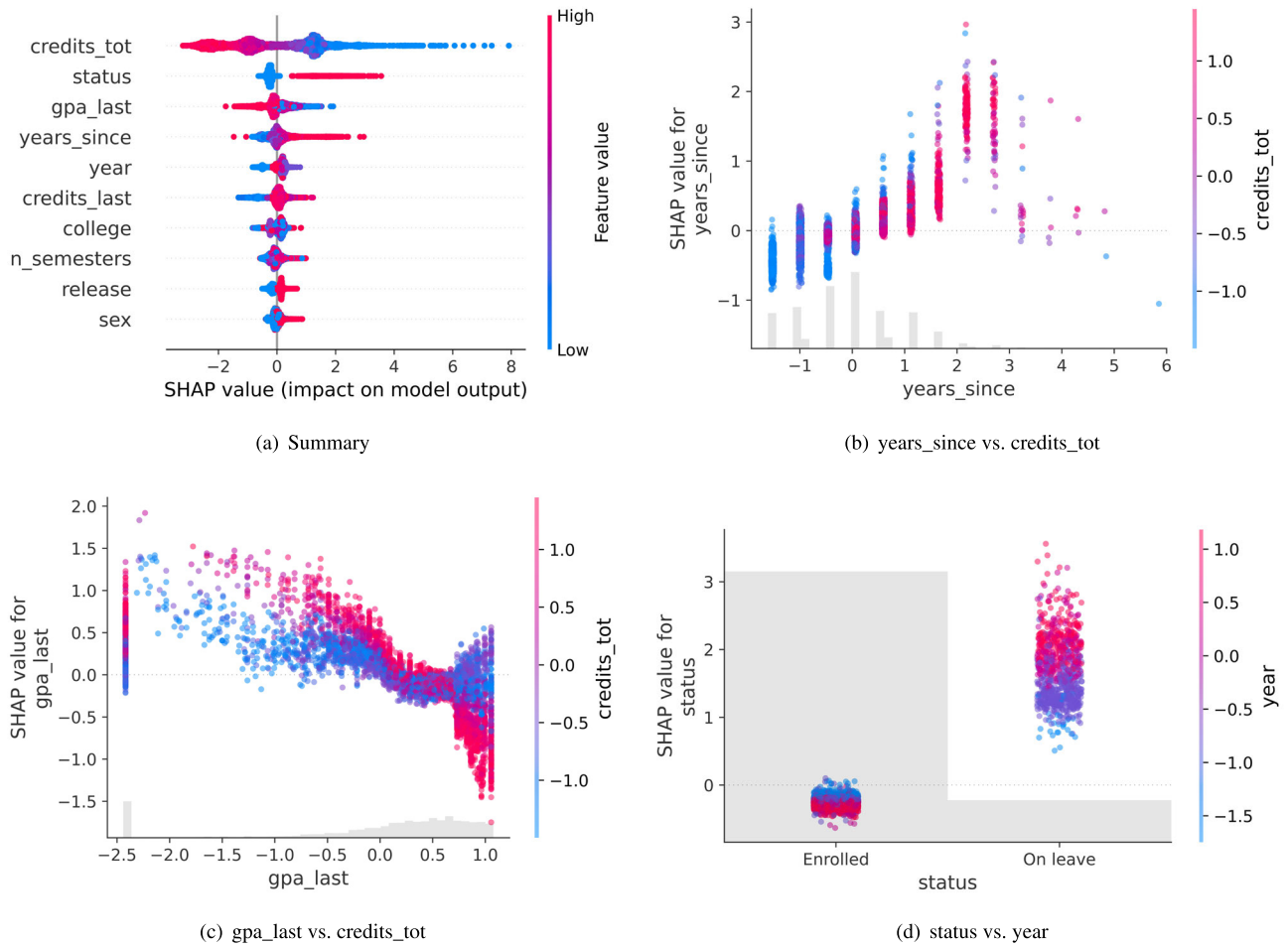
The three *baseline* models (no resampling) achieve Brier scores between 0.0337 and 0.0351, already indicative of good calibration. Among them, CatBoost hugs the diagonal most closely, whereas Stacking shows noticeable zigzags. After resampling, all CatBoost variants shift *below* the diagonal—an overconfidence bias that grows with the degree of class-ratio distortion (see Table 9). CatBoost+Tomek records nearly the same Brier score (0.0353) as the baseline Stacking model, yet its curve still sits below the diagonal; CatBoost+SMOTE is the most overconfident of all.

To estimate the calibration curve without reusing training data, we split the held-out 20% test set into two equal halves: the first half served as a calibration subset for fitting the Platt logistic curve, and the second half was retained for final evaluation. The resampled curves now lie much closer to the diagonal, confirming that post-hoc scaling can recover well-calibrated probabilities. Baseline CatBoost and XGBoost, already near-perfect, improve only marginally, whereas the effect is substantial for CatBoost+SMOTE, CatBoost+ENN, and CatBoost+Tomek.

SHAP assigns each feature in each sample a contribution value—the *SHAP value*—whose sign and magnitude indicate how much that feature pushes the prediction upward or downward [50]. We computed SHAP values for the baseline CatBoost model with `shap 0.46`.

Figure 3(a) summarizes the ten most influential features; importance is the mean absolute SHAP value across the test set. `credits_tot` dominates, followed by `status`, `gpa_last`, and `years_since`. Fewer accumulated credits, a leave of absence, a low last-semester GPA, and many years since admission all raise the dropout risk.

Panel (b) is a SHAP dependence plot for `years_since` (x-axis) with SHAP(`years_since`) on the y-axis; points are



(a) Summary

(b) years_since vs. credits_tot

(c) gpa_last vs. credits_tot

(d) status vs. year

FIGURE 3. SHAP analysis for CatBoost (baseline) at $t = 2$. Panels (b)–(d) are SHAP dependence plots: the x-axis shows the raw feature value and the y-axis shows the feature's SHAP value; points are colored by the auxiliary feature indicated by the right-hand color bar (z-score standardized). Color encodes the standardized auxiliary feature value (blue = low, red = high). Histograms under the x-axes show feature distributions.

colored by standardized credits_tot (red = many, blue = few). Although higher years_since generally increases risk, the effect is modulated by credits: students who have stayed long but earned few credits receive the largest positive SHAP values. Once years_since exceeds roughly two standard deviations, the interaction with credits diminishes.

Panel (c) plots gpa_last (x-axis) against SHAP(gpa_last) (y-axis), again colored by credits_tot. A higher GPA usually lowers dropout probability, but the decline is much steeper for students with many credits (red). Students with comparatively few accumulated credits—often in their early years—retain positive SHAP values even at high gpa_last, consistent with early-stage withdrawals or transfers.

Panel (d) contrasts status (enrolled vs. on leave); points are colored by standardized academic year (red = senior, blue = freshman). Being on leave strongly increases risk (positive SHAP values), with the effect largest for later-year students—cases less likely to reflect routine military leave and more likely to indicate disengagement.

In addition to the global patterns above, the same SHAP machinery can be applied to a *single* prediction, yielding a ranked list of features that pushed the student's score up or down. The resulting explanation—e.g., "Risk 0.81: −24 cumulative credits vs. expected progress (major driver), last-semester GPA −0.7, current status = on leave"—gives advisors clear, case-specific talking points and suggested levers for intervention. We therefore can export these local explanations alongside every alert produced by the calibrated CatBoost model.

While global SHAP summaries reveal overall trends, local explanations are computed for every prediction, highlighting the most influential variables for each individual student. For clarity, Table 11 presents a concise input–output example demonstrating how these features are transformed into calibrated dropout probabilities, binary alerts, and interpretable explanations.

Calibration converts raw model scores into *actionable* risk probabilities, while SHAP explains *why* a student is flagged. Together these post-hoc tools turn an accurate yet opaque classifier into a decision-support instrument that advisors can *trust* (well-calibrated) and *understand* (feature-level rationale). This stage therefore closes the final research

TABLE 11. Illustrative input–output examples of the proposed framework.

Inputs (subset)				Outputs		
gpa_last	credits_tot	status	···	Risk	Alert	Top SHAP drivers[a]
2.10	24	on leave	···	0.71	1	years_since, status, ...
3.60	98	enrolled	···	0.18	0	credits_tot, gpa_last, ...

[a]Each driver may be summarized either by its SHAP contribution or by its actual feature value (e.g., deviation from cohort mean).

gap by making early-warning predictions both reliable and interpretable—prerequisites for any real-world intervention program.

IV. DISCUSSION

The experiments in Section III show that the proposed framework provides competitive dropout prediction with practical lead time and high interpretability. This section interprets those results through the lens of the four research gaps highlighted earlier and distills operational lessons for practitioners.

A. ADDRESSING THE FOUR RESEARCH GAPS

1) ACTIONABLE HORIZONS AND COHORT COVERAGE

Early-warning systems must balance timeliness against accuracy. By framing each semester snapshot as the start of a rolling t-semester window, we (i) remove the traditional first-year-only constraint and (ii) guarantee a non-zero prediction horizon. Empirically, performance declined as t increased, yet the drop from one to two semesters was smaller than the drop from two to three. A two-semester horizon therefore emerges as a pragmatic choice: it grants advisors roughly six months to one year to intervene while still preserving robust predictive power. Because the rolling-window logic operates on semester indices rather than absolute dates, institutions on quarter or trimester calendars can adopt the same procedure by adjusting t accordingly. In this way, the framework simultaneously closes Gap 1 (limited horizon) and Gap 2 (cohort restriction).

2) CLASS IMBALANCE IN PRACTICE

Dropout data are typically *extremely* skewed (e.g., about 5% positives in our case), which makes accuracy and ROC–AUC poor metrics. We therefore deploy two complementary remedies:

- Algorithm-level tuning — After the model is trained, we sweep the decision *threshold* from 0 to 1 on cross-validated predictions and keep the value that maximizes F1.[1] This single step alone can convert a high-precision/low-recall model into one that captures many more true dropouts with only a modest loss of precision.
- Data-level rebalancing — We then apply oversampling and undersampling techniques to the *training split only*. After every such resample, the model is *refitted* and the decision threshold *retuned* on the validation split; the configuration that yields the highest validation F1 is carried forward to the held-out test set.

Because undersampling is inexpensive and non-destructive, we recommend it as a low-risk default, especially when practitioners prefer simpler calibration pipelines with minimal distribution shift, thereby addressing Gap 3 (inadequate imbalance handling).

[1]The threshold is a *post-training* parameter, not a model hyperparameter, so it can be tuned without refitting.

3) CALIBRATION AND EXPLANATION

Even a highly ranked model will be ignored if advisors cannot trust its output. Platt scaling turns raw scores into well-calibrated, interpretable probabilities while adding negligible computational overhead. SHAP analysis then translates each student-level prediction into a ranked list of feature contributions—actionable evidence that advisors can communicate to the student (“Your cumulative credits are far below expected for your year,” “A recent leave of absence increases your risk,” etc.). The combination satisfies both halves of Gap 4 (scarce calibration and explainability): numerical reliability and transparent rationale.

Pilot feedback from academic advisors at the case-study university indicated that such labeled, evidence-rich alerts were more likely to trigger timely outreach than opaque classifier scores. Informal reactions during early demonstrations further revealed that calibrated probabilities were substantially easier for advisors to interpret than raw model outputs, and that SHAP-based explanations helped them contextualize risk factors during student meetings. Collectively, these impressions suggest that transparent, evidence-backed alerts are perceived as more actionable and trustworthy in real advising contexts.

B. COMPARISON WITH RECENT FULL-PIPELINE STUDIES

Among recent works, Song et al. [23] and Liu et al. [31] are most closely aligned with the present study in scope, as both aimed to develop end-to-end dropout-prediction pipelines. However, their designs differ substantially from ours across key methodological dimensions related to cohort coverage, prediction horizon, class imbalance handling, and calibration with explainability.

Song et al. [23] extended dropout prediction to all academic years and compared multiple feature-table generation strategies, including a final-semester representation that effectively performs post-hoc classification immediately before dropout or graduation. Although longitudinal data were used, each student’s multi-semester record was aggregated into a single feature table rather than organized in a rolling structure, resulting in a limited prediction horizon and reduced applicability for proactive early-warning. Class imbalance was handled only through a single-step SMOTE procedure, and neither probability calibration nor explainability was incorporated. In contrast, our framework maintains the longitudinal sequence of student records within rolling multi-semester windows, explicitly defines a non-zero

horizon, integrates threshold optimization with resampling, and embeds calibrated and interpretable outputs through Platt scaling and SHAP analysis.

Liu et al. [31] used administrative data from 4,424 first-year students at a Portuguese university to develop a three-class model (Graduate, Dropout, Enrolled). Because the dataset contained only first-year information, the model effectively predicts outcomes at the end of the first academic year and cannot utilize updated information for upper-year students, thereby limiting cohort coverage. Moreover, the labeling time of the three classes was not explicitly defined, leaving the prediction horizon ambiguous and inconsistent across samples. The dataset exhibited low imbalance, and no resampling or calibration was applied, although SHAP analysis was performed for post-hoc interpretation. Consequently, Liu et al.'s work represents a dataset-specific model rather than a generalizable framework. Our approach, by contrast, defines a clear temporal horizon, incorporates imbalance-handling and calibration mechanisms, and provides institution-independent portability through modular design.

Overall, these comparisons highlight that while recent full-pipeline studies have made important progress toward integrating modeling and interpretability, they remain limited by static designs and dataset-specific constraints. The proposed framework advances beyond these boundaries by establishing a dynamic, generalizable, and operational early-warning system that is portable across institutions and sustainable for real-world advising contexts.

C. OPERATIONAL DEPLOYMENT AND INTERVENTION STRATEGY

To translate predictive insights into administrative action, we designed a three-stage operational workflow for applying this framework at the target institution.

(1) Periodic Screening: Upon the release of the semester snapshot (e.g., April 1st for Fall, October 1st for Spring), the system executes the inference pipeline using the newly updated registry data. It generates a calibrated risk probability for every enrolled student. Administrators can then define a ''priority intervention group'' by filtering students whose risk scores exceed the optimized decision threshold (e.g., > 0.54) or by ranking them to allocate limited counseling resources effectively.

(2) Diagnostic Counseling: Unlike ''black-box'' risk lists, the framework supports precision consulting by providing student-level SHAP explanations alongside every alert. As illustrated in the input–output example in Table 11, advisors receive not just a risk probability but a ranked list of contributing factors (e.g., ''Risk 0.81 driven by Low Cumulative Credits (-24 vs. expected) and Recent GPA Drop''). This enables advisors to understand why a specific student is at risk and discuss targeted remedies—such as course retaking plans to recover credits or financial aid applications to address scholarship loss—rather than offering generic advice.

(3) Institutional Planning: Beyond individual support, the framework allows the administration to aggregate SHAP values to identify systemic risk drivers. For instance, if recent academic performance (gpa_last) consistently appears as a top risk factor for a specific college, the university can review curriculum difficulty or assign tutoring resources. Similarly, if trends in scholarship-related features (e.g., loss of funding due to GPA cutoffs) emerge as primary drivers of attrition, the administration can realign financial planning to distribute funds where they most effectively prevent dropout.

From a technical standpoint, for the CatBoost implementation, this full refresh typically completes in under an hour on a standard server (16 vCPU, 64 GB RAM), keeping the early-warning system continuously aligned with the most recent student data. Student data must be rigorously de-identified before any analysis begins. In this study all personally identifying information was stripped at the extraction stage, ensuring that only anonymized records entered the modeling pipeline, while the deployment system securely re-links predictions to student IDs for authorized advising. Institutions adopting the framework should enforce the same upfront anonymization and still give students a clear opt-out option.

From an operational standpoint, the calibrated risk scores and threshold-based flags produced by the pipeline can be directly incorporated into existing university systems such as student information systems or advising dashboards. By automatically generating high-risk flags and providing SHAP-based explanations during consultations, the framework enables departments to prioritize limited intervention resources based on consistent and interpretable risk indicators.

D. LIMITATIONS AND FUTURE WORK

Our empirical evaluation used data from a single university to demonstrate the framework's end-to-end feasibility. Nevertheless, its generalizability stems from a design that relies only on generic registry data and off-the-shelf algorithms, rather than on multi-site evidence. Thus, multi-institution replication is not required to claim design-level universality, but it remains valuable to quantify external performance and portability across diverse student populations [51]. In this sense, the framework is generalizable by design, but its external validity must still be empirically verified.

Two extensions appear most promising. First, future work should pursue multi-institution replication to assess portability across different educational contexts. Second, while this study focuses on standard tabular learners to prioritize operational simplicity, the framework is modular by design. Future extensions could incorporate more complex architectures—such as sequential models (e.g., LSTM, Transformers) or survival analysis formulations—to further contextualize predictive performance.

Beyond technical validation, causal follow-up studies are needed to test whether advisor interventions, triggered by the

calibrated/SHAP-explained model, actually reduce dropout rates. Previous studies have demonstrated that predictive analytics can inform actionable student-success interventions such as early-alert and advising systems [52], [53]. These follow-up studies will constitute the real-world validation phase, during which the framework will be deployed in live advising environments to evaluate its operational effectiveness and the measurable impact of model-guided interventions on student retention.

Finally, recent studies have emphasized that model evaluation should include fairness auditing in addition to accuracy [54]. Because models trained on historical registry data may inadvertently reflect systemic inequalities embedded in academic records, there is a risk that predictions could reinforce existing disparities. To mitigate this, our framework's emphasis on probability calibration and SHAP-based interpretability provides a practical foundation for fairness auditing (e.g., by checking calibration across demographic subgroups). Building on this perspective, future work will incorporate periodic fairness audits and ethical-use guidelines to ensure that model predictions support students equitably and transparently across demographic groups. Beyond fairness, responsible deployment requires vigilant monitoring of false positives, as incorrect alerts can drain institutional resources. Crucially, even well-intentioned interventions can inadvertently harm students if "at-risk" labels trigger stereotype threat or stigmatization; thus, risk scores should be communicated carefully and transparently. To this end, predictions should serve strictly as decision-support signals rather than automated decision outcomes, helping ensure that predictive analytics augments—rather than replaces—human judgment in advising.

In summary, the discussion confirms that each stage of the framework systematically mitigates a specific research gap, while the end-to-end pipeline remains lightweight, portable, and operationally feasible for real-world deployment by data-limited institutions.

V. CONCLUSION

We presented a generalizable, end-to-end framework for long-term student-dropout prediction that addresses four persistent gaps by covering all academic years via a rolling t-semester design and guaranteeing actionable lead time. In experiments, gradient-boosting models—especially CatBoost—paired with post-training threshold optimization performed robustly under severe imbalance; light undersampling (ENN, Tomek-links) yielded small but consistent gains, whereas SMOTE-type oversampling gave mixed results when train–validation class ratios diverged. Platt scaling produced well-calibrated probabilities, and SHAP provided transparent, feature-level rationale, turning accurate yet opaque scores into advisor-ready alerts. The pipeline is lightweight and portable, requiring only standard registry data and minimal customization. Future work will evaluate portability across universities and assess, via randomized or quasi-experimental studies, whether calibrated, SHAP-guided outreach actually reduces dropout.

DATA AND CODE AVAILABILITY

The student-level administrative records are owned by Sun Moon University and cannot be publicly released due to institutional and national privacy regulations. Qualified researchers may request controlled access from the corresponding author with the university's permission. All Jupyter notebooks and implementation scripts used in this study are available at: https://github.com/jbkwon11/dropout_prediction.

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